Verification and Validation Report

Numerical correctness, convergence and NREL UAE Phase VI comparison

Software	AeroBlade 2.1.3
Document	AB-VVR-2.1.3
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AeroBlade 2.1.3 is distributed as a Windows portable application. Verify the SHA-256 checksum before scientific use.

Document control

2.1.3	2 August 2026	Windows CPython 3.14.2 validation rerun	Version 2.1.3

Contents

Executive summary

AeroBlade 2.1.3 passed the defined source-level verification and validation acceptance matrix. The Windows rerun completed 16/16 unit tests, exact physical-identity checks, interpolation/quadrature/optimizer/CAD checks, radial and iterative sensitivity studies, and an external rotor-power comparison for one NREL UAE Phase VI configuration. All radial stations converged in the reported cases.

Principal external result. With the Himmelskamp rotational-augmentation option, Phase VI power MAPE was 12.159%, RMSE 1.203 kW and R^2 0.851 over 5-10 m/s. This satisfies the predeclared MAPE limit of 15% but does not validate every operating regime.

Scope limitation. The external comparison covers one axial, zero-yaw, fixed-speed/fixed-pitch rotor configuration and rotor-integrated power/torque. It does not validate yaw, dynamic stall, turbulent inflow, local pressure distributions, wake dynamics, tower interference, aeroelasticity, structural design, noise or certification loads.

1. Configuration under test

Item	Recorded configuration
Software	AeroBlade 2.1.3 release-candidate source
Validation platform	3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]; win32
Validation timestamp	2026-08-02T06:24:32Z
Duration	70.19 s
Tests	16 unittest functions plus independent validation harness
Rotor benchmark	NREL UAE Phase VI, S809, zero yaw, 72 rpm, 3° pitch, 5-10 m/s
Experimental-use rule	No model coefficient fitted to the experimental power table
Packaged release gate	Same package must record PASS in BUILD-INFO.txt and RELEASE-MANIFEST.json

The portable builder creates a clean environment, compiles the source, reruns the numerical suite, builds the one-folder executable, removes/disallows Qt Virtual Keyboard, performs an off-screen GUI smoke test, starts bundled XFOIL and writes component hashes. The V&V claims in this report concern the numerical code and listed configuration; the package manifest is the release-specific deployment evidence.

2. Verification and validation method

Verification asks whether the implemented equations and algorithms are solved correctly. Validation asks whether the chosen mathematical model agrees sufficiently with physical observations for a declared use. The campaign therefore separates code/equation tests, solution verification and external comparison.

Static/runtime	compileall; 16 unit tests; packaged smoke gate	Detect import, syntax, resource and regression failures.
Equation identities	P = Q omega; Cp = lambda Cq; Betz bound	Check dimensional and non-dimensional consistency.
Component algorithms	Viterna boundaries; Reynolds interpolation; Weibull; optimizer	Check isolated numerical implementations against exact/analytic references.
Solution verification	Radial-grid convergence; iteration tolerance/relaxation	Quantify numerical sensitivity of integrated power.
External validation	NREL Phase VI power/torque	Assess one physical rotor configuration without coefficient fitting.

Geometry output	STL facet/edge/volume audit	Check closed-manifold aerodynamic surface generation.
Reproduction	Linux reference versus Windows rerun	Detect platform-dependent numerical drift.

3. Acceptance matrix

Compilation	Return code 0	PASS	0
Unit suite	All 16 pass	PASS	16/16
Power identity	max |P-Qω| <= 1e-10 W	PASS	0.0e+00 W
Coefficient identity	max |Cp-λCq| <= 1e-12	PASS	5.55e-17
Physical power coefficient	max Cp < 0.593	PASS	0.409616
Phase VI Himmelskamp	Power MAPE <= 15%	PASS	12.159%
Radial discretization	80+ stations: max error <= 1% vs 640	PASS	0.800%
Iteration sensitivity	max power difference <= 0.5%	PASS	0.0655%
Reynolds interpolation	max exact-case error <= 1e-12	PASS	8.88e-16
Weibull quadrature	mean relative error <= 1e-12	PASS	1.18e-16
Optimizer	distance to exact solution <= 1e-4	PASS	7.51e-07
STL topology	0 boundary and 0 non-manifold edges	PASS	0 / 0
Cross-platform rerun	all recorded scalars agree within 1e-12	PASS	maximum observed difference 0

4. Code and equation verification

4.1 Unit and compilation suite

The Windows harness ran Python byte-code compilation and unittest discovery independently of the GUI. The 16 functions cover runtime paths/version resources, polar interpolation and extrapolation, BEM identities and corrections, design equations, energy quadrature, optimization and CAD topology. Both compilation and all tests passed.

4.2 Physical identities and solver convergence

- Maximum |P - Qω|: 0.0e+00 W.
- Maximum |Cp - λCq|: 5.551e-17.
- Maximum predicted Cp: 0.409616, below 0.593.
- All station solutions converged for the physical-identity, mesh, iteration and Phase VI cases.

4.3 Viterna-Corrigan implementation

The corrected two-sided implementation preserves measured source coefficients exactly, matches positive and negative outer boundaries independently and applies the rear-flow symmetry transform. Boundary jumps evaluated 1e-6 degree outside the measured interval remained of order 1e-8 or smaller.

Quantity	Observed
Maximum source-coefficient change	0.00e+00
Positive-side Cl/Cd jumps	4.03e-09 / 1.31e-08

Quantity	Observed
Negative-side Cl/Cd jumps	9.83e-09 / 1.44e-08
Rear-flow Cl/Cd symmetry error	0.00e+00 / 0.00e+00
Cd at 90 degrees	1.308

5. NREL UAE Phase VI external comparison

The benchmark uses the two-bladed, tapered/twisted S809 NREL UAE Phase VI rotor at zero yaw, 72 rpm and 3 degrees pitch. Geometry is represented by 19 published aerodynamic stations. Predictions were generated before the published rotor-level power/torque table was read; no correction coefficient was fitted to those values.

Data provenance. Geometry follows NREL/TP-500-29955 Appendix A. The six rotor-level comparison values used here were transcribed from the comparison table published by Al-Ttowi et al. (Processes 12(9), 1994, 2024), not re-extracted from the raw NREL measurement archive. Future validation should replace or cross-check this secondary transcription with campaign data and uncertainty metadata.

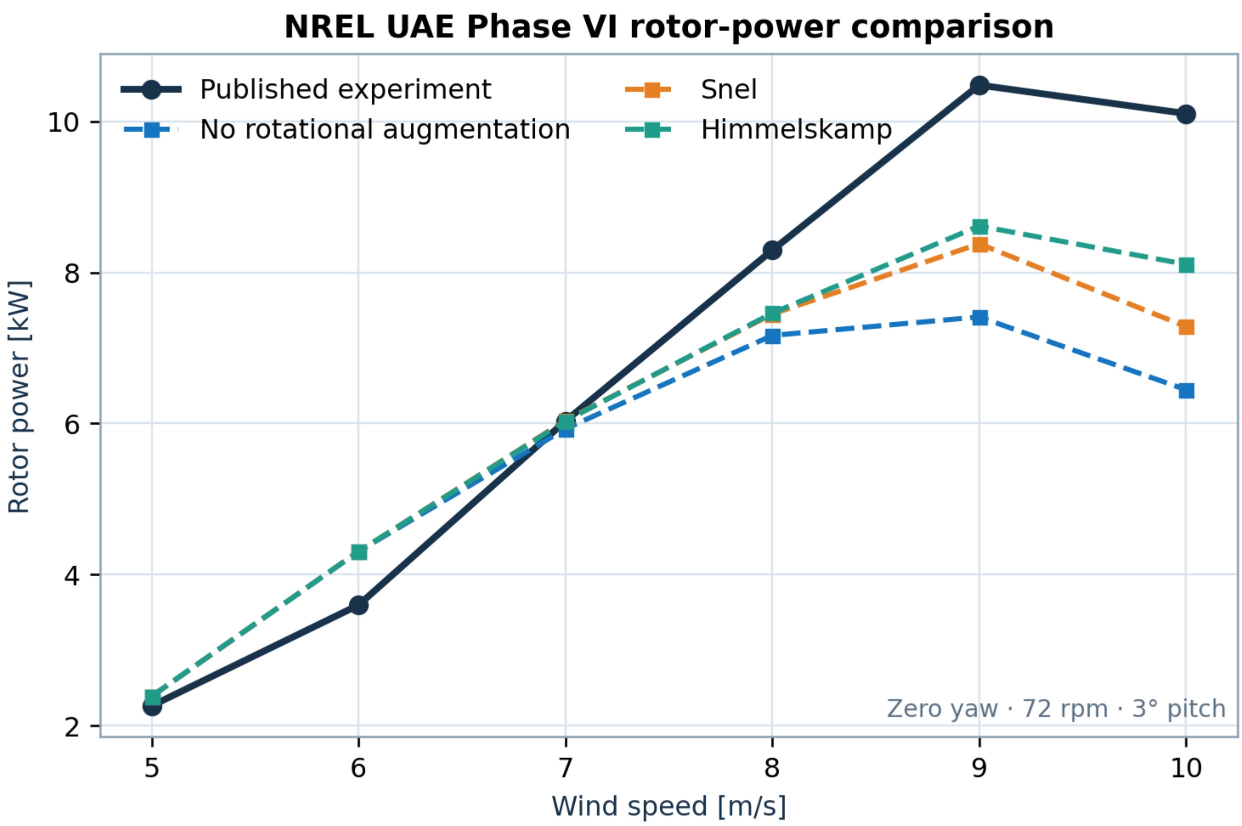

Figure 1. Published rotor power and AeroBlade predictions for three rotational-augmentation choices.

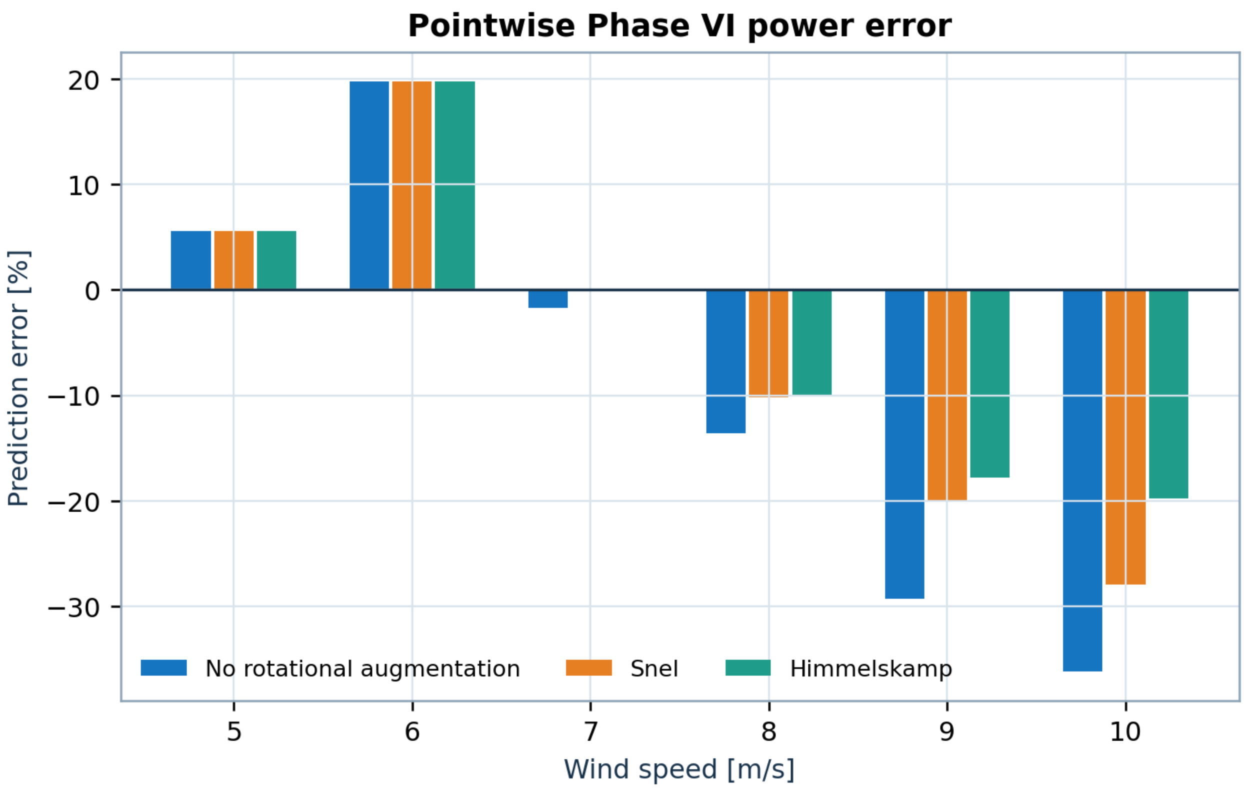

Figure 2. Pointwise relative error, showing increasing underprediction in deep stall.

5.1 Himmelskamp pointwise results

5	2.262	2.389	+5.610	0.3927	100%
6	3.596	4.306	+19.737	0.4096	100%
7	6.032	6.032	+0.002	0.3614	100%
8	8.294	7.461	-10.048	0.2994	100%
9	10.480	8.615	-17.800	0.2428	100%
10	10.103	8.107	-19.758	0.1666	100%

5.2 Aggregate model sensitivity

None	-1.187	1.466	2.025	17.690	0.577
Snel	-0.820	1.102	1.506	13.942	0.766
Himmelskamp	-0.643	0.922	1.203	12.159	0.851

Himmelskamp gives the best aggregate agreement in this limited set. That ranking is a model-sensitivity observation, not universal calibration evidence. Errors change sign: slight overprediction at 5-6 m/s becomes underprediction at 8-10 m/s, consistent with the limits of steady 2-D-polar BEM in stalled three-dimensional flow.

6. Numerical solution verification

6.1 Radial discretization

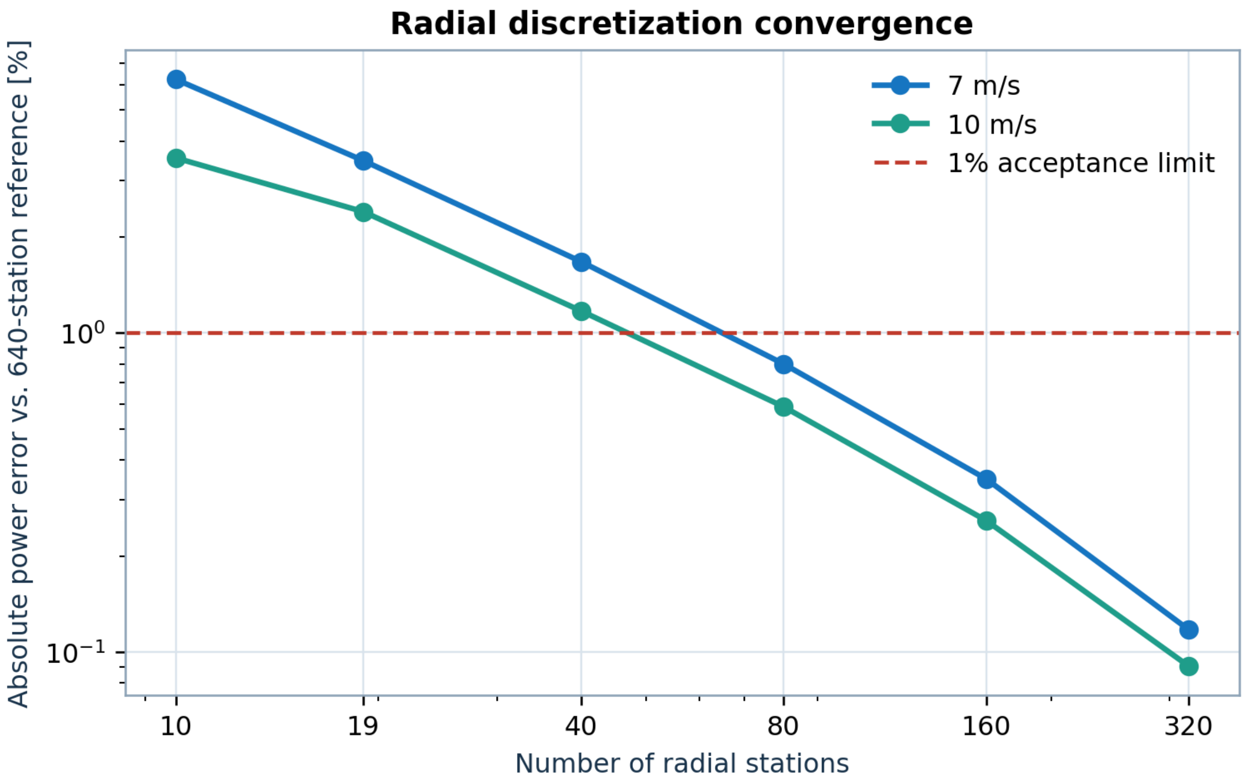

Figure 3. Integrated-power convergence relative to a 640-station reference.

At 80 stations, the maximum absolute error across 7 and 10 m/s is 0.800%, satisfying the 1% criterion. The 19-station geometric definition is an input table, not a statement that 19 numerical elements are always adequate. Scientific calculations should report and justify the discretized station count.

6.2 Iterative tolerance and relaxation

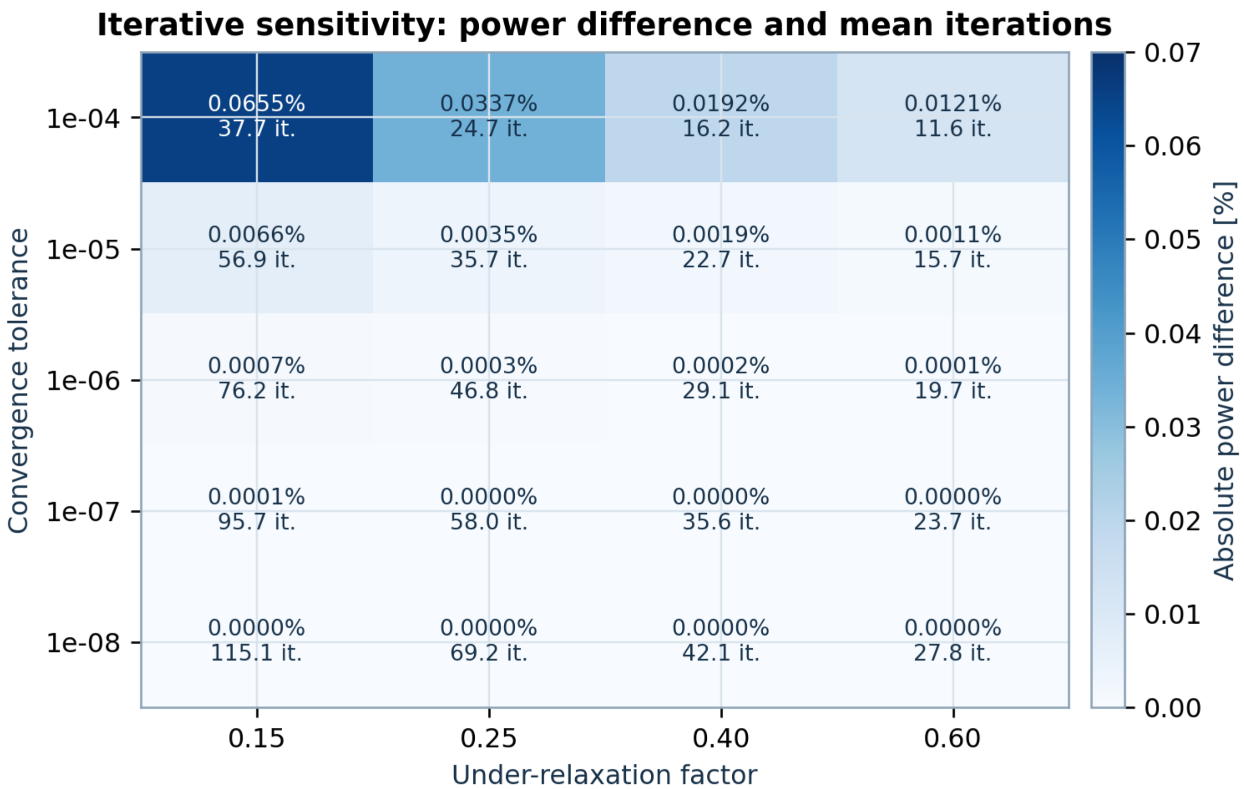

Figure 4. Power sensitivity and mean iteration count across tolerance/relaxation settings.

Across tolerances 1e-4 to 1e-8 and relaxation factors 0.15 to 0.60, every station converged and the maximum power difference to the reference was 0.0655%. Tighter tolerances reduce power drift but can increase iterations; reporting only convergence without parameter sensitivity is insufficient.

7. Component and output verification

Reynolds interpolation	Exact synthetic interpolation cases	max error 8.88e-16
Weibull quadrature	Analytic mean $A\,\Gamma(1+1/k)$	relative error 1.18e-16; PDF integral 0.9999999977
Optimizer	Quadratic exact optimum (0.25, 0.25)	distance 7.51e-07; 55 iterations
STL topology	Every mesh edge shared exactly twice	3040 facets; 0 boundary; 0 non-manifold
STL geometry	Positive area and non-zero enclosed volume	min area 8.272e-05 m^2; $\lvert V\rvert$ 0.098762 m^3

8. Cross-platform reproducibility

The complete validation harness was executed on Linux for the archived supplement and rerun on Windows 64-bit with standard CPython 3.14.2. CSV outputs and JSON metrics compared exactly for all recorded numerical values at absolute/relative tolerance 1e-12; the maximum observed difference was zero. This is evidence of deterministic reproduction for these cases, not a guarantee across every processor, dependency version or user input.

9. Uncertainty, limitations and validity domain

- Experimental uncertainty bars were not available in the secondary rotor-power table used here; error metrics therefore treat reported values as exact.
- The six-point data set is small and concentrated at one fixed-speed/fixed-pitch, zero-yaw condition.
- S809 polar fidelity, rotational augmentation and steady BEM assumptions dominate deep-stall discrepancies.
- Rotor-integrated power can hide compensating spanwise load errors; local pressure/load validation is still required.
- No wake-field, acoustic, structural, fatigue, controller or aeroelastic evidence is claimed.
- The Himmelskamp result should not be generalized to other blades without independent validation.
- The STL audit establishes topological closure only, not manufacturability or structural adequacy.

Recommended use. Use AeroBlade 2.1.3 for preliminary steady HAWT aerodynamic studies, method teaching and controlled sensitivity analyses when the user documents inputs and performs case-specific convergence and validation checks.

10. Release acceptance and traceability

1. Source compilation and the 16-test numerical suite must pass in the clean build environment.
2. The packaged GUI must start and close under --smoke-test.
3. The bundled XFOIL executable must start successfully.
4. Qt Virtual Keyboard files must be absent from the proprietary distribution.
5. The project license, citation file, user guide, this V&V report, third-party notices, XFOIL GPL text/source and dependency inventory must be present.
6. AeroBlade.exe must report product/file version 2.1.3 and the release ZIP must pass its SHA-256 and ZIP-integrity checks.
7. BUILD-INFO.txt and RELEASE-MANIFEST.json in the same extracted package are the final evidence that these deployment gates passed.

10.1 Traceability to artifacts

Evidence	Release artifact
Numerical outputs	validation_summary.json and CSV tables in the V&V supplement
Test transcript	unit_tests.log and builder log
Phase VI input data	nrel_phase_vi_geometry.csv, nrel_phase_vi_experimental.csv and S809 polar file
Package dependency record	DEPENDENCIES.txt and LICENSES/THIRD-PARTY-LICENSES.txt
Component integrity	RELEASE-MANIFEST.json
Archive integrity	AeroBlade-2.1.3-Windows-x64-Portable.zip.sha256

11. Reproduction procedure

From a clean copy of the validation supplement, place or select the AeroBlade 2.1.3 source root and execute validate_aeroblade.py using standard 64-bit CPython 3.11-3.14 with the declared runtime dependencies. The harness reads independent CSV inputs, generates results in a new output directory and ends with VALIDATION PASS only if every acceptance rule succeeds.

```
python validate_aeroblade.py --source <AeroBlade-2.1.3> --output <new-results-dir>
```

For the native portable release, run tools/build_windows.py with the selected standard CPython interpreter. Preserve the build log, manifest, dependency inventory and ZIP checksum with the archived release.

12. References

1. Hand, M. M. et al. Unsteady Aerodynamics Experiment Phase VI: Wind Tunnel Test Configurations and Available Data Campaigns. NREL/TP-500-29955, 2001. DOI: 10.2172/15000240.
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4. Prandtl, L. Applications of Modern Hydrodynamics to Aeronautics. NACA Report 116, 1921.
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